FRAMEWORK FOR DATA AND VISUAL ANALYTICS

MINI PROJECT

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Abstract:

The Sales Performance Analysis and Dashboard project is designed to study sales data and present useful business insights through data visualization tools such as Microsoft Power BI and Tableau. The dataset used contains 10,000 sales records with fields such as order ID, order date, region, city, product category, quantity, sales amount, profit, order status, sales channel, and payment method.

The dataset is loaded into both Power BI and Tableau to build interactive and informative visualizations. In Power BI, key performance indicators (KPIs) including Total Sales, Total Profit, Total Orders, and Total Quantity are calculated and shown through card visuals. Additional charts are built to examine sales by region, monthly sales trends, product categories, and order status. In the same way, Tableau is used to create a dashboard made up of four visualizations: Sales by Region, Monthly Sales Trend, Sales by Category, and Order Status.

The dashboards allow users to recognize sales patterns, compare how regions perform, find the most profitable categories, and review how order statuses are distributed. By turning raw sales data into graphical form, the project makes the information simpler to understand and supports data-driven decision-making.

This project shows the practical use of business intelligence and data visualization techniques in sales analysis. The finished dashboards can help business users keep track of performance, spot trends, and make well-informed decisions that improve sales and profitability.

Objectives:

1. To analyze a sales dataset of 10,000 records using Power BI and Tableau.
2. To compute key performance indicators such as total sales, total profit, total orders, and total quantity.
3. To show regional sales performance using bar charts.
4. To study monthly sales trends using line charts.
5. To compare sales across product categories using pie or donut charts.
6. To present the distribution of order statuses through graphical visualizations.
7. To build clear and user-friendly dashboards that make data easy to interpret.
8. To support business decision-making by identifying sales patterns and performance trends.

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Implementation:

Dataset Collection:

A sales dataset of 10,000 records is gathered in CSV format. It holds details about orders, dates, regions, products, quantities, sales, profits, and order statuses.

Data Import:

The CSV file is loaded into Microsoft Power BI and Tableau Desktop. The data fields are reviewed to confirm that they are suitable for analysis and visualization.

Data Preparation:

The dataset is inspected for correct column types and overall usability. Date, numeric, and categorical fields are identified so that the required visualizations can be created.

Dashboard Development in Power BI

The following measures are defined using DAX:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity

KPI values are displayed using card visuals. A bar chart compares sales across regions, a line chart shows monthly sales trends, and donut charts represent sales by category and by order status.

Dashboard Development in Tableau:

The same sales dataset is connected to Tableau Desktop, and four worksheets are built:

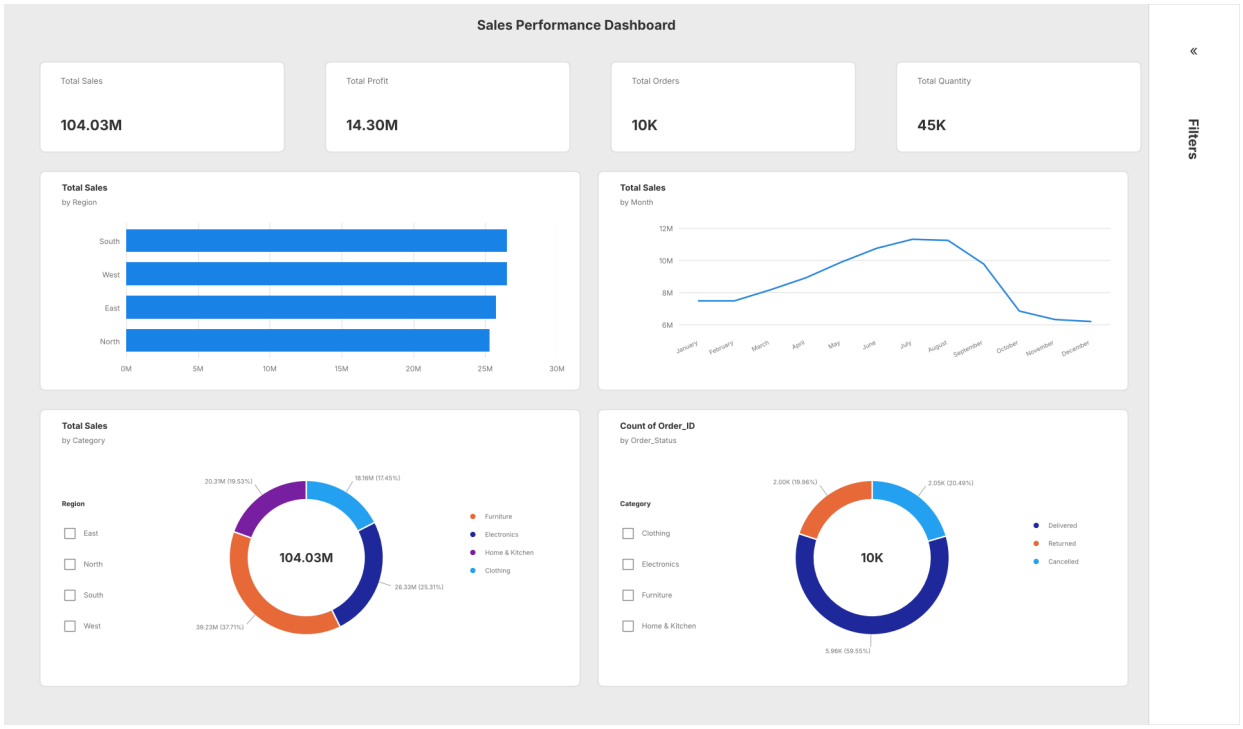
- Sales by Region — a bar chart that compares sales across regions.
- Monthly Sales Trend — a line chart that shows how sales change over time.
- Sales by Category — a pie chart that represents how sales are split across product categories.
- Order Status — a pie chart that shows the distribution of orders by status.

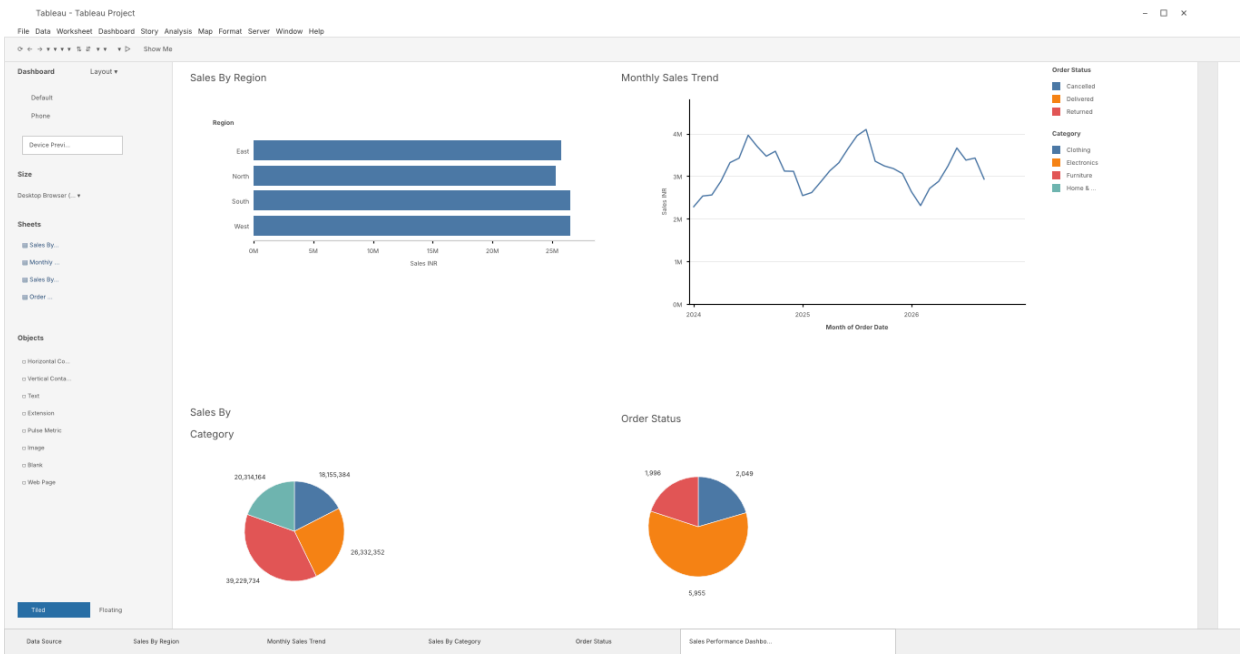
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Output:





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Conclusion:

The Sales Performance Analysis and Dashboard project clearly shows how Microsoft Power BI and Tableau can turn raw sales data into meaningful visual information. Working with a dataset of 10,000 records, the project presents the main sales indicators and visualizes regional performance, monthly sales trends, product category distribution, and order status.

The Power BI dashboard gives a summary of key performance indicators together with several charts, whereas the Tableau dashboard brings four worksheets into a single visual reporting interface. These dashboards make sales information easier to understand and help users identify patterns and compare business performance.

The project offers hands-on experience in data importing, data preparation, measure calculation, chart creation, and dashboard development. In the future, the system could be extended with interactive filters, profit margin analysis, year-over-year comparisons, and sales forecasting techniques. Overall, the project highlights the importance of data visualization in business intelligence and data-driven decision-making.